Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/12/2017
PI Name:	Venkateswaran, Kasthuri Ph.D.		
Project Title:	ISS Microbial Observatory - a Genetic Approach		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Organization Name:	NASA Jet Propulsion Laboratory		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/01/2013	**End Date:**	09/30/2018
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS Microbial Observatory NOTE: Extended to 9/30/2018 per F. Hernandez/ARC (Ed., 3/26/18) NOTE: Extended to 12/31/2017 per F. Hernandez/ARC (Ed., 8/22/17) NOTE: Extended to 04/30/2017 per F. Hernandez/ARC (Ed., 2/2/17) NOTE: Extended to 12/31/2016 per F. Hernandez/ARC (Ed., 4/11/16) NOTE: Extended to 4/30/2016 per A. Chu/ARC (Ed., 9/17/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fox, George (University of Houston) Pierson, Duane (NASA Johnson Space Center)		
Grant/Contract No.:	19-12829-26 under Task Order NNN13D111T		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As recommended by the National Research Council Decadal Survey, the Jet Propulsion Laboratory (JPL) proposes to establish an International Space Station-Microbial Observatory (ISS-MO) to generate a microbial census of the space station's surfaces and atmosphere using advanced molecular microbial community analysis techniques, supported by traditional culture-based methods and modern bioinformatic computational modeling. The proposed ISS-MO establishment will lead to long-term, multigenerational studies of microbial population dynamics, addressing recommendation P1 of this solicitation. It will also provide answers to NASA Space Biology Guiding Questions CMM-1, CMM-4, and CMM-5. The ISS-MO project's methodology will serve as the foundation for an extensive microbial census, offering significant insight into spaceflight-induced changes in the populations of beneficial and potentially harmful microbes. Additionally, it will provide NASA with both a mechanistic understanding of these changes (e.g., cataloging population changes and mapping/linking these to environmental niche and genomic changes), as well as insight into practical countermeasures for mitigating risks to humans and environmental systems. The ISS-MO team will use existing ISS sample collection technologies to generate an initial microbial census. Following their return to Earth, samples from the various ISS modules will be analyzed using standardized technologies from the Mars Program-funded projects. The proposed ISS-MO project includes the delivery of a database that compiles the genomic sequences and genetic information for all the microbes encountered within the ISS habitat. Using this data, NASA can more accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to providing microbial profiles, the ISS-MO team will identify which microbial taxa pose particular threats to crew health. Furthermore, specific ISS-MO project aims will enable us to resolve applicable NASA-Human Research Program integrated research plan risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The NASA crewmembers are exposed constantly to the dust in a closed system, and have also reported allergies. Subsequent microbiological analyses using traditional methods noticed no anomaly and were also not able to predict the causative microbial agents. The next-generation sequencing study provided NASA with the ability to assess the spectrum of microorganisms associated with the ISS surfaces and would help to predict possible microbial agents that might affect crew health. The results of the present study also exhibited presence of representative human associated (skin, gastrointestinal track, wounds, etc.) bacterial commensals and opportunistic fungal pathogens. Furthermore, based on the phylogenetic affiliation it is possible to determine whether these viable microorganisms are hardy and robust that have been previously reported to survive in oligotrophic environments (cleanrooms, hospital, commercial airliner, etc.). The study has improved our understanding of background microbial contamination, thus facilitating the development of biosensors to monitor closed habitats like ISS and future manned missions.
Task Progress:	The importance of cataloguing the ISS microbiome, which consists of both culturable and unculturable microbes, parallels the reasons prompting the surge in research into the “built microbiome” here on Earth. Emerging studies on the microbiome of homes, offices, classrooms, museums, and hospitals have revealed an assemblage of bacteria, fungi, viruses, and protozoa unique to that indoor environment that are influenced by a variety of factors such as building design, ventilation, humidity, air pressure and flow, occupant numbers or activities performed. Specific microbes in these indoor spaces have been shown to impact human health such as influencing our susceptibility to allergies, infectious diseases, or sick building syndrome. The influence of the indoor microbiome on human health becomes more important for astronauts during flight missions due to lowered immunity as a result of spaceflight and the lack of sophisticated medical interventions that are available on Earth. In light of an upcoming new era of human expansion in the universe, such as future space travel to Mars, the microbiome of the closed space environment needs to be examined thoroughly to identify the types of microorganisms that can accumulate in the space environment, their persistence, survivability and significance to human health and impact on spacecraft infrastructure. For this reason, the National Research Council (NRC) Decadal Survey recommended NASA to establish a coordinated, large-scale Microbial Observatory program within the ISS platform. As part of this NASA initiative, the microbial communities on ISS surfaces from 8 different locations over three Flight missions, spanning 14 months, were characterized using culture-based techniques, qPCR and iTag sequencing of the 16S rRNA gene and internal transcribed spacer (ITS) region using the Illumina platform. PMA-treatment was carried out (before subjecting to various methodologies) to distinguish dead and intact/viable cells within microbial populations. This comprehensive analysis of the ISS microbiome was used to assess how microbial communities change over time (temporal distribution) and throughout the ISS locations (spatial distribution). The implementation of novel molecular techniques to monitor intact microbial populations of the ISS environment opens a possibility for broadening the current surveillance practices to maintain the health of the crew and to promote advance in deep space human habitation in the future.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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